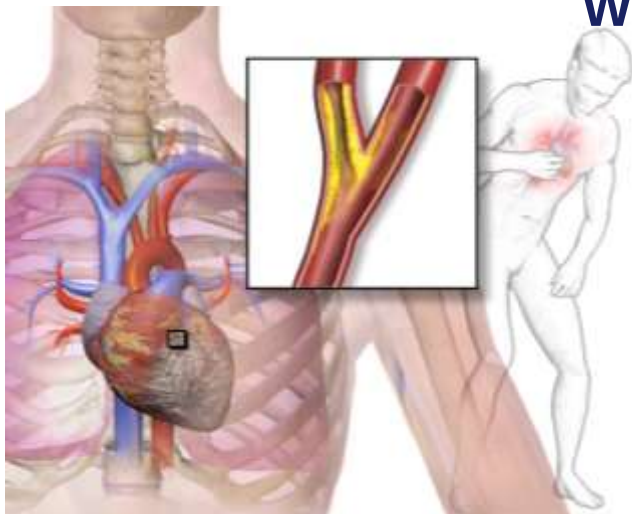


Pharmacology of Antianginal Agents



Warisara Parichatikanond, Ph.D., Dr.Scient.Med.

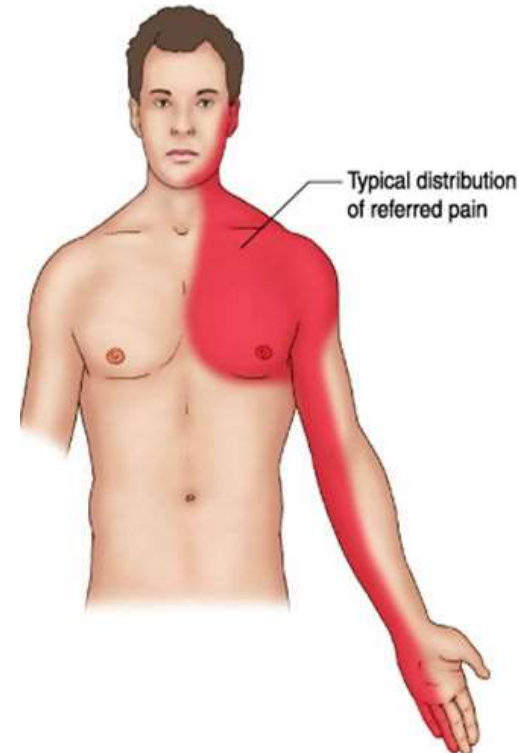
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Angina pectoris

- Angina pectoris is the principal symptom of **ischemic heart disease (IHD)**
- Angina pectoris: **Chest pain** results from myocardial ischemia (MI), is characterized by **sudden, severe substernal pain or pressure**. Chest pain **radiate** to arms and neck.
- Primary cause: An **imbalance** between myocardial **oxygen demand & oxygen supply** by coronary vessels
 - ↓ myocardial oxygen supply
 - ↑ myocardial oxygen demand
 - Or both

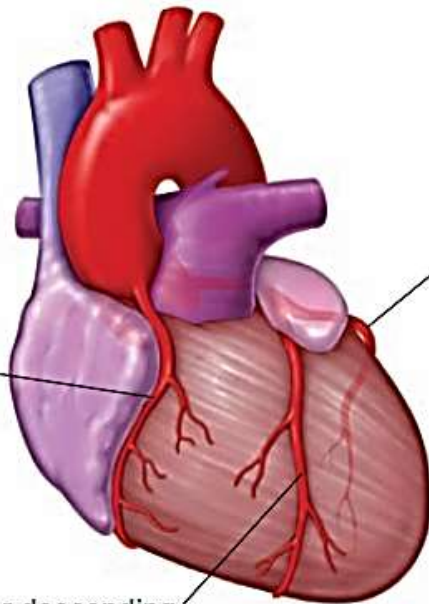


Myocardial O₂ supply & demand

Myocardial O₂ supply

Perfusion of the heart

Vascular tone of the coronary arteries



Left circumflex coronary artery

Right coronary artery

Left anterior descending coronary artery

Myocardial O₂ demand

Ventricular wall stress

Diastolic

Systolic

Preload

Afterload

Venous tone

Arteriolar tone

Heart (pump)

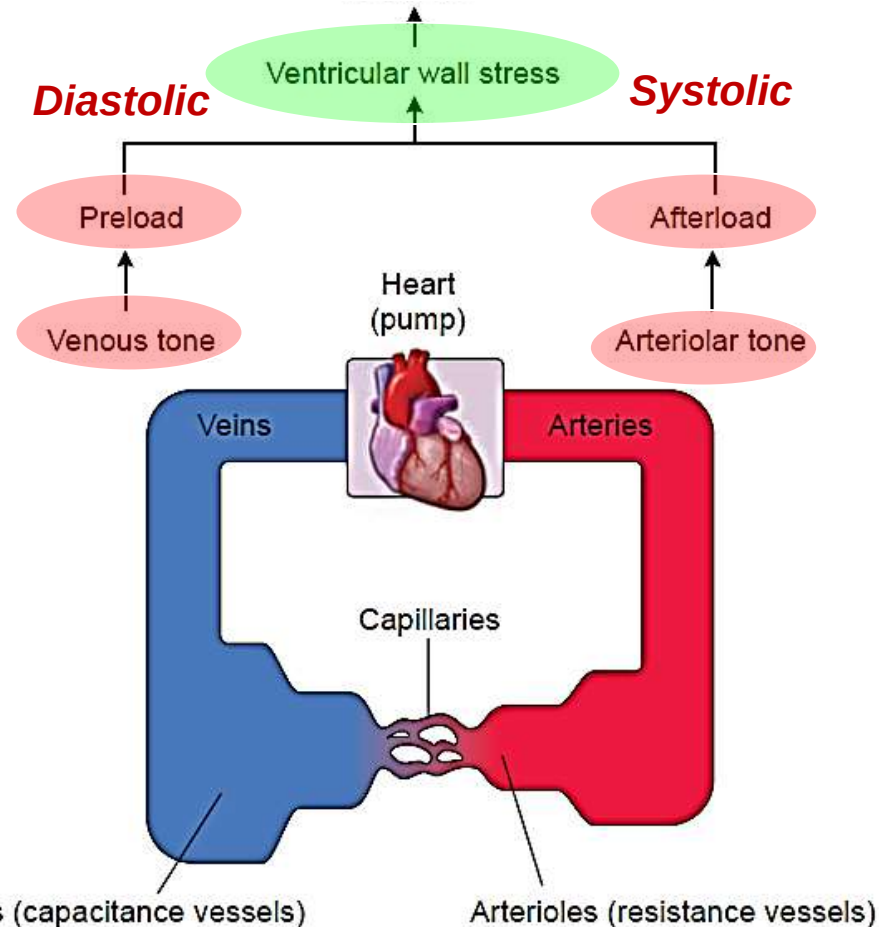
Veins

Arteries

Capillaries

Veins (capacitance vessels)

Arterioles (resistance vessels)



Pathophysiology of angina

- **Determinants of myocardial O₂ demand**
 - **Wall stress:** Intraventricular pressure, ventricular volume, wall thickness
 - **Heart rate** (chronotropic state)
 - **Contractility** (inotropic state)
 - **Myocardial metabolism** (glucose VS fatty acid)
- **Determinants of myocardial O₂ supply: Coronary blood flow****
 - **Perfusion pressure** (aortic diastolic pressure) and duration of diastole
 - **Coronary vascular resistance** (intrinsic & extrinsic factors)
- **Determinants of vascular tone**
 - **Arteriolar tone** -> Systolic wall stress
 - **Venous tone** -> Diastolic wall stress

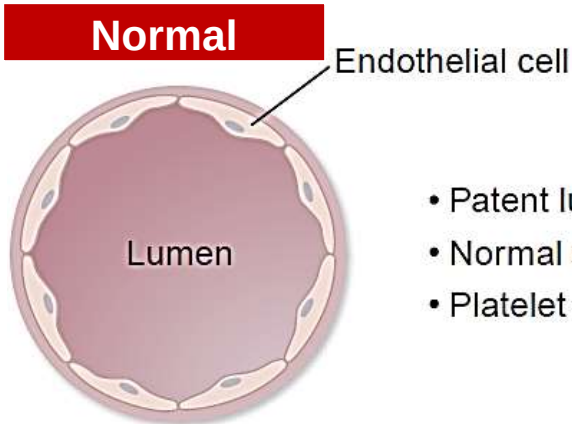
Types of angina

- **Stable angina (classic angina, chronic stable angina):** Most common
 - **Angina of effort**:** No symptom at rest (chest pain during exercise, stress)
 - Cause: **Atherosclerotic plaque**
 - Symptoms: **Predictable**, moderate pain and short (< 5 mins), **relieved by rest or medication** (such as NTG sublingual)
- **Unstable angina:** Less common
 - **Angina at rest**:** Symptom occurs while at rest
 - Cause: **Atherosclerotic plaque + plaque rupture**** → platelet activation → thrombus → blood clot
 - Symptoms: **Unpredictable**, more severe and prolong than stable angina
- **Variant angina (vasospastic angina, Prinzmetal's angina):** 2% of patient
 - Cause: **Vasospasm** (hypercontractility of vascular smooth muscle)
 - Symptoms: Very painful, usually occur between midnight and morning

Pathophysiology of anginal syndromes

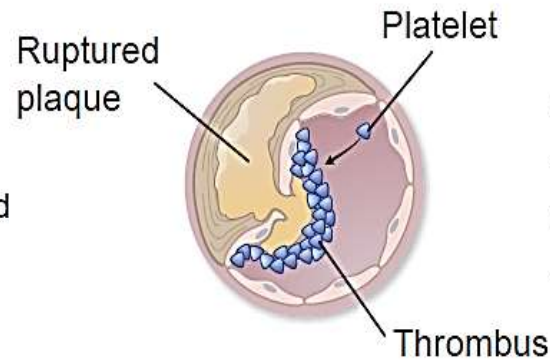
Coronary flow reduction

Normal



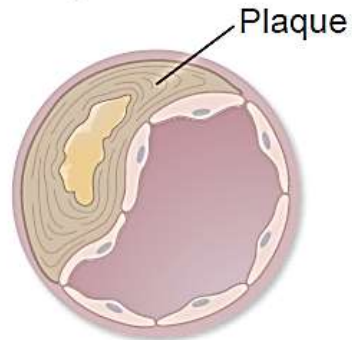
- Patent lumen
- Normal endothelial function
- Platelet aggregation inhibited

Unstable angina



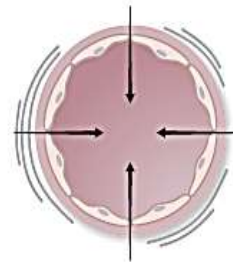
- Plaque ruptured
- Platelet aggregation
- Thrombus formation
- Unopposed vasoconstriction

Stable angina



- Lumen narrowed by plaque
- Inappropriate vasoconstriction

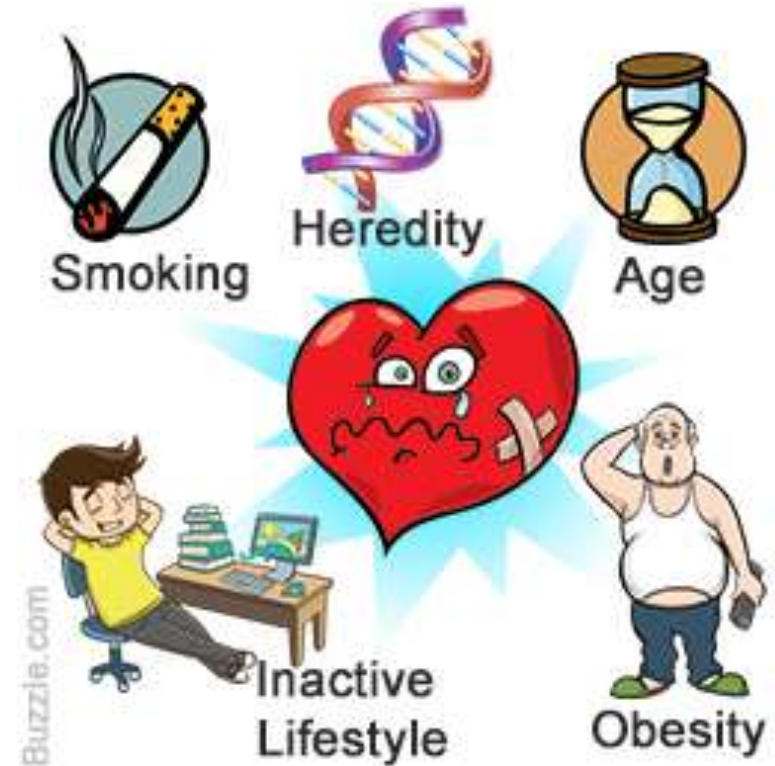
Variant angina



- No overt plaques
- Intense vasospasm

Risk factors of angina

- Age (older people)
- Cigarette smoking
- Diabetes mellitus (DM)
- Dyslipidemia
- Hypertension
- Obesity (BMI ≥ 30 kg/m²)
- Physical inactivity
- Family history of cardiovascular diseases



Treatment of Angina Pectoris

Non-pharmacologic approaches

Pharmacologic approaches

Surgery



Treatment of angina pectoris

- **Non-pharmacologic approaches: Life-style modifications**

- **Pharmacologic approaches:**

- **Antianginal drugs:**

- Organic nitrates
 - Calcium channel blockers (CCBs)
 - Beta-blockers
 - Other antianginal drugs



General Mechanism:

↓ Myocardial oxygen consumption

- ↓ Preload and/or afterload

- ↓ Heart rate

↑ Myocardial blood & oxygen supply

- Dilate coronary artery or arterioles

- **Co-therapy in angina -> Long-term vascular protection**

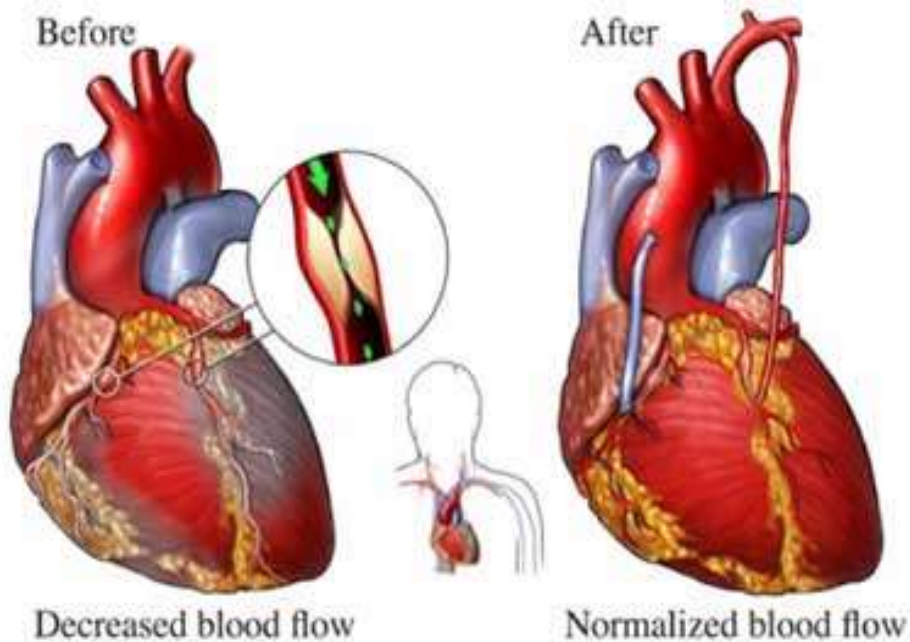
- Antithrombotic: Aspirin and/or clopidogrel
 - Antihyperlipidemic: Statins
 - Antihypertensive: ARBs or ACEIs (Protect from MI)

- **Combination therapy**

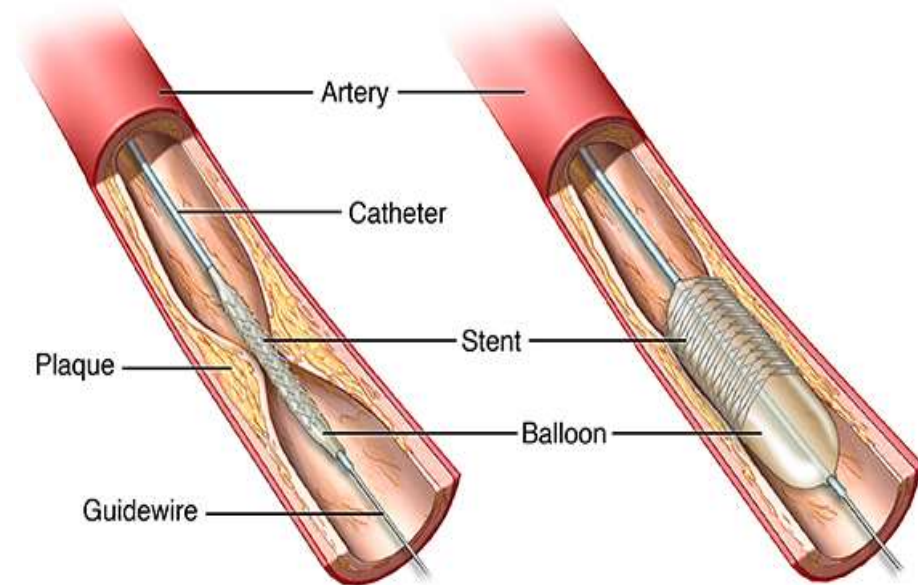
- **Surgery**

- Coronary artery bypass graft (CABG)
 - Percutaneous transluminal coronary angioplasty (PTCA or PCA)

Surgery of coronary artery



**Coronary artery bypass graft
(CABG)**



**Percutaneous transluminal coronary
angioplasty (PTCA or PCA)**

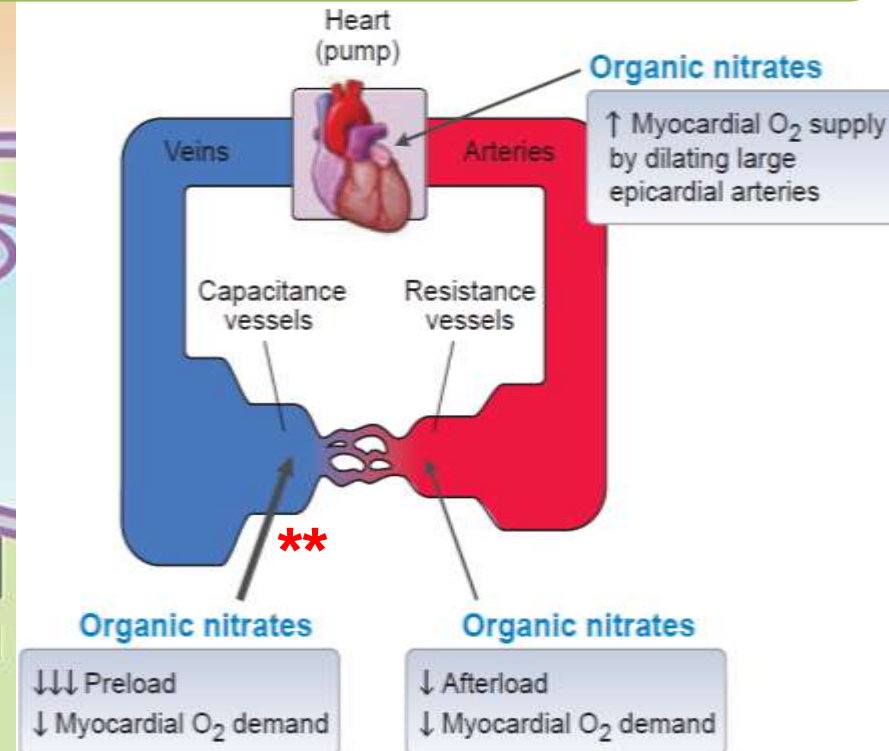
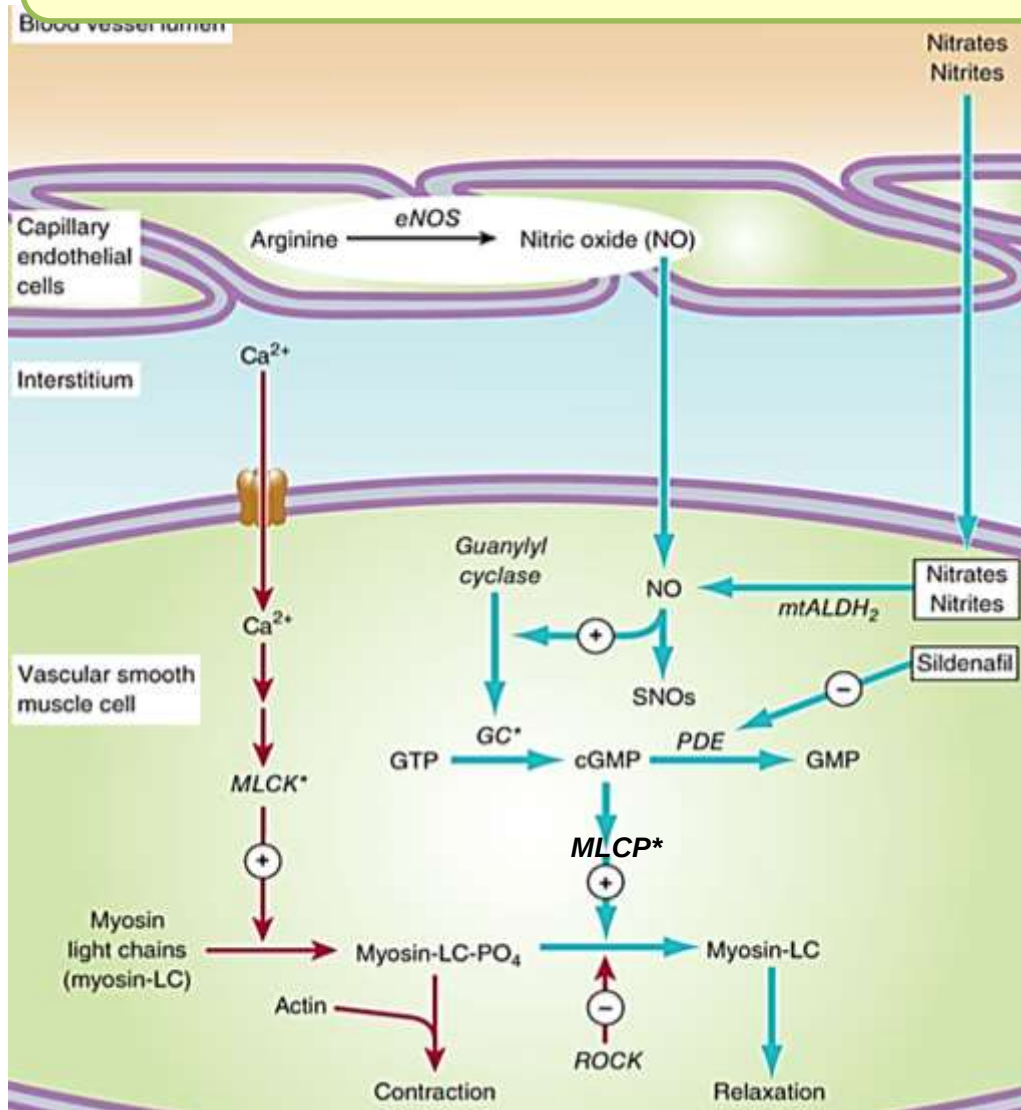
Organic nitrates

Nitroglycerin (NTG)

Isosorbide dinitrate (ISDN)

Isosorbide 5-mononitrate (5-ISMN)

Organic nitrates: Mechanisms & Clinical effects



↓ myocardial O₂ demand

- **Major effect: Dilate venous*** (→ ↓ Preload, ↓ ventricular end-diastolic pressure)
- **Minor effect (large dose): Dilate arterial** (→ ↓ PVR, ↓ Afterload)

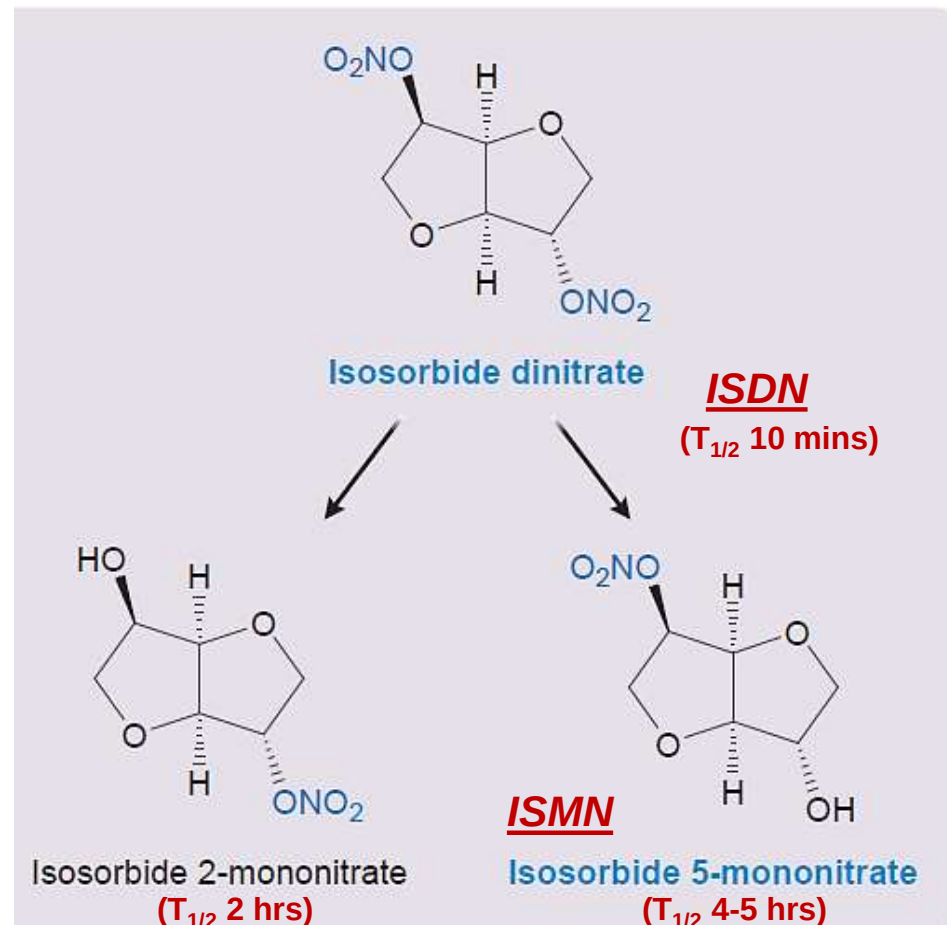
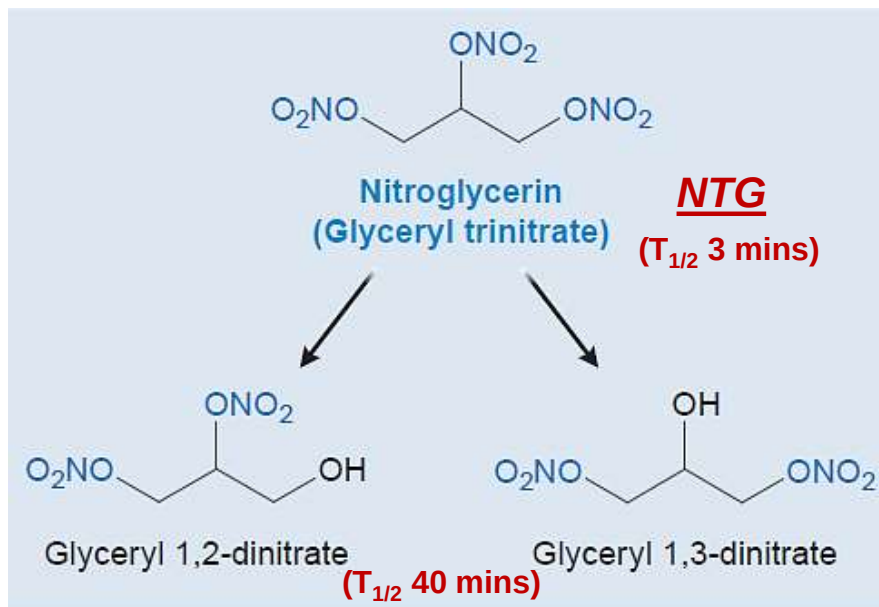
↑ coronary blood flow:

- ↑ blood supply to ischemia area

↑ collateral blood flow

↑ NO → Activate guanylate cyclase → ↑ cGMP

Organic nitrates: Structures & metabolism



NTG and ISDN are biologically active nitrates that are metabolized into **active molecules** with **longer half-lives** than their parent compounds.

Organic nitrates: Pharmacokinetics

- **NTG and ISDN:**
 - **High hepatic first pass elimination:** Oral bioavailability is very low
 - **Sublingual or transdermal** -> Avoids the first-pass effect
- **5-ISMN:**
 - **Less hepatic first pass metabolism:** Oral bioavailability 100% -> **Oral form**

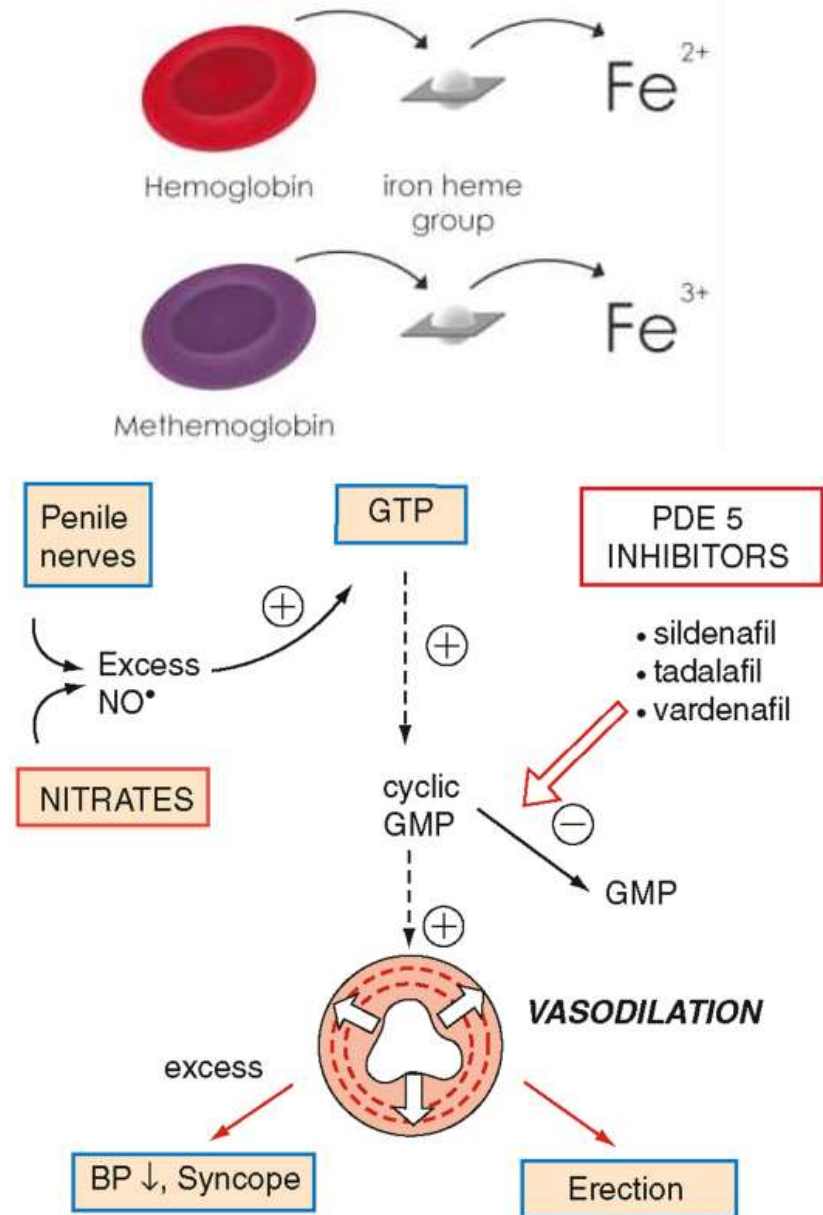
<u>Property</u>	<u>NTG</u>	<u>ISDN</u>	<u>5-ISMN</u>
Half-life (min)	3	10	280
Plasma clearance (L/min)	50	4	0.1
Apparent volume of distribution (L/kg)	3	4	0.6
Oral bioavailability (%)	< 1	20	100

Route of administration:

- **Sublingual: NTG and ISDN:** Rapid onset (1-3 min), short duration (20-30 min)
-> **** For acute therapy ****
- **IV : Nitroglycerin:** Treat severe recurrent unstable angina
- **Slowly absorbed preparations (oral, buccal, transdermal): ** For maintenance therapy****
(6-10 hrs), but can lead to “nitrate tolerance”

Organic nitrates: Adverse effects & cautions

- **Adverse effects:**
 - **Orthostatic hypotension**, syncope
 - **Reflex tachycardia**
 - Headache, dizziness, flushing
 - **Methemoglobinemia** (rarely but serious)
- **Nitrate tolerance:**
 - Continuous or frequent exposure
 - Solution: **Nitrate-free periods** (~ 8 hrs)
 - Tablet: TID, night- free
 - Transdermal patch (24-hr): Remove patch during the night
- **Caution:**
 - **PDE-5 inhibitors**
(sildenafil, tadalafil, vardenafil)



Organic nitrates: Clinical uses

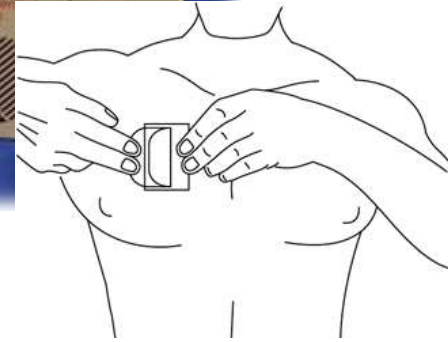
Clinical uses: All types of angina, acute MI, CHF

Drug	Dose	Duration
Short-acting	For acute angina attack (injection, sublingual, spray)	
Nitroglycerin, SL	0.15 - 1.2 mg	10 - 30 minutes
Isosorbide dinitrate, SL	2.5 - 5 mg	10 - 60 minutes
Long-acting	For angina prophylaxis (sustained release, slow release, transdermal)	
Nitroglycerin, oral sustained-action	6.5 - 13 mg per 6 - 8 hours	6 - 8 hours
Nitroglycerin, slow-release, buccal	1 - 2 mg per 4 hours	3 - 6 hours
Nitroglycerin, slow-release, transdermal	10 - 25 mg per 24 hours (one patch per day)	8 - 10 hours
Isosorbide dinitrate, oral	10 - 60 mg per 4 - 6 hours	4 - 6 hours
Isosorbide mononitrate, oral	20 mg per 12 hours	6 - 10 hours
Pentaerythritol tetranitrate (PETN), oral	50 mg per 12 hours	10 - 12 hours



ISDN, sublingual (SL)

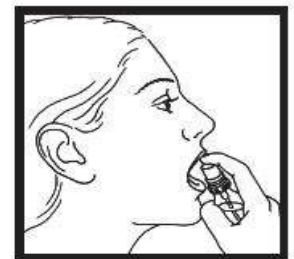
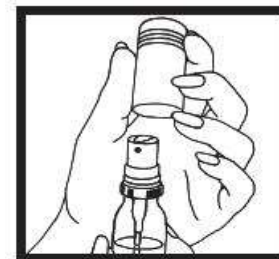
Use at the first sign of chest pain. Sit stable. Place the tablet under tongue and allow it to dissolve slowly. (Do not swallow). Drug should be effective within 3 mins, **if symptom does not improve within 5 min, repeat 1 more tab.** If chest pain persists after a total of 3 tabs -> Go hospital!



NTG, slow-released, transdermal patch

Dose, number of doses per day, time interval, and length of time depends on individual. Apply one patch once a day in the morning. Leave the patch in place for a total of 12 to 14 hours. **Schedule nitrate-free period of time every day!**

NTG, oral spray



Calcium channel blockers (CCBs)

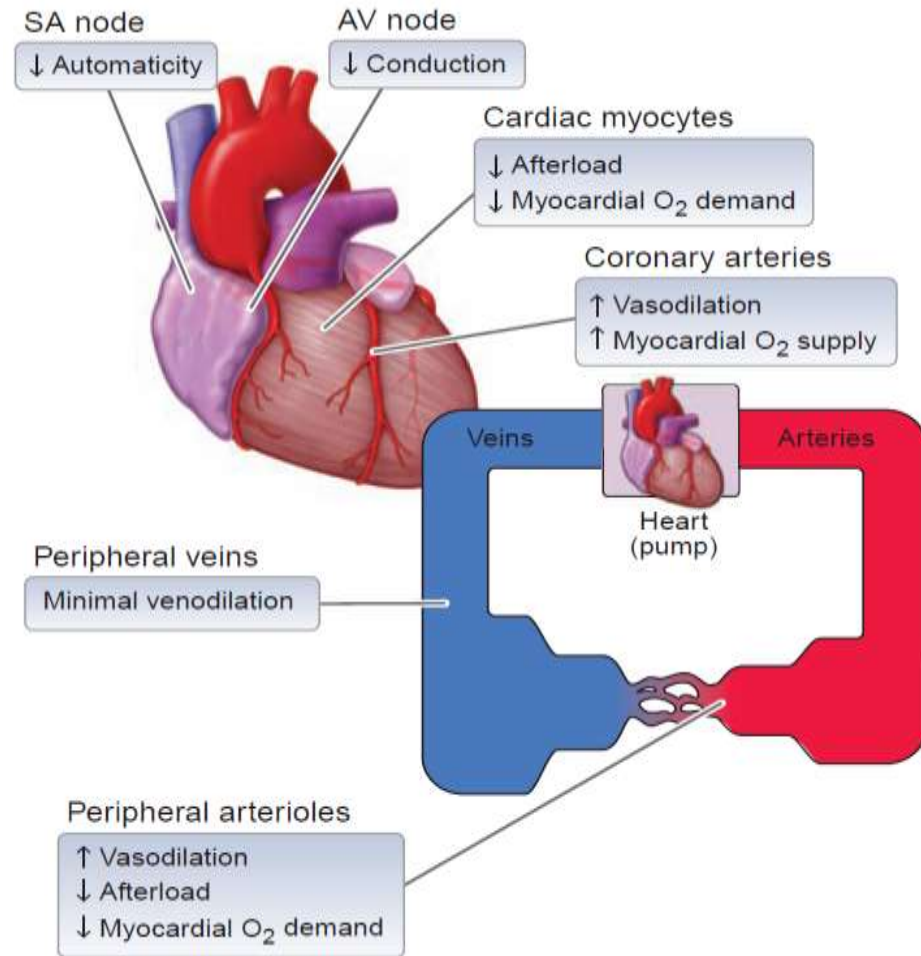
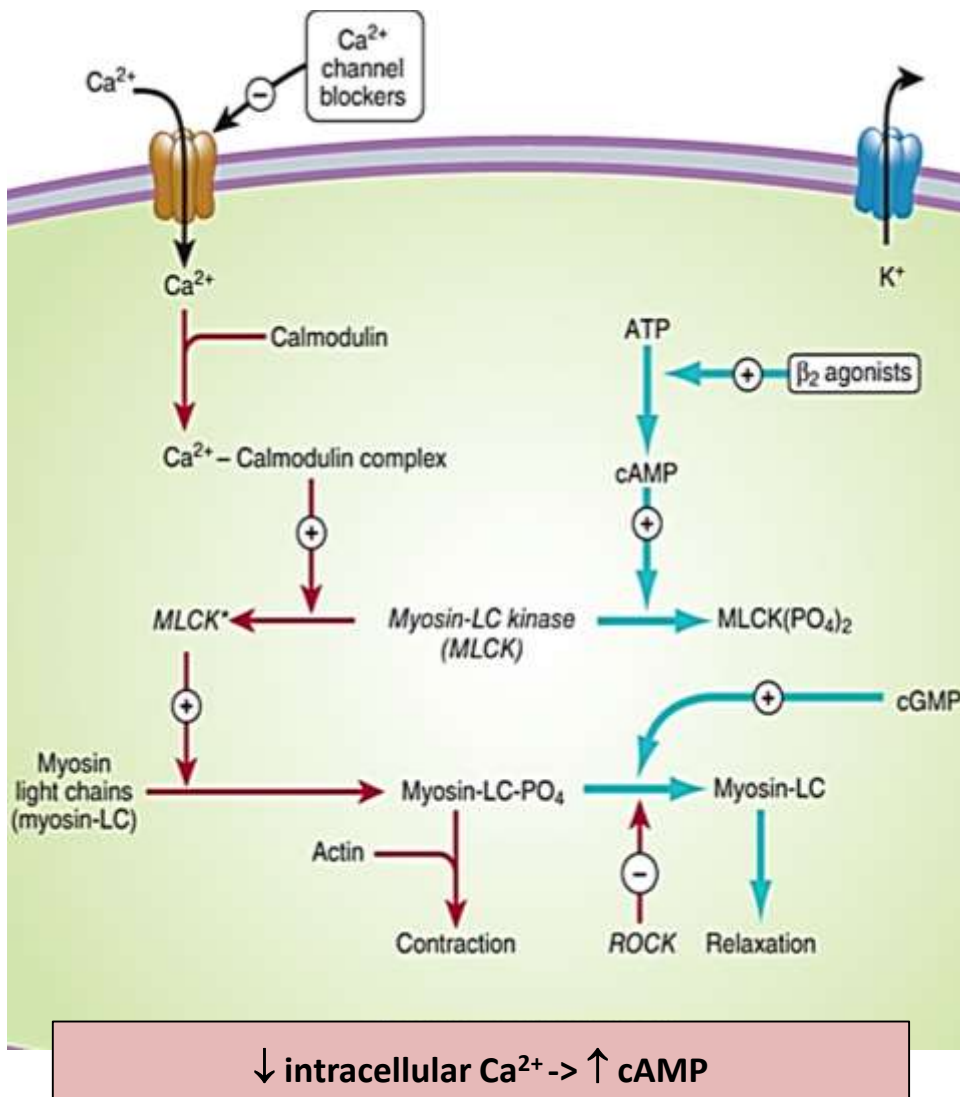
DHPs:

Amlodipine, nifedipine,
felodipine, nicardipine, etc.

Non-DHPs:

Diltiazem
Verapamil

CCBs: Mechanisms & Clinical effects



- CCBs **dilate coronary arteries and peripheral arterioles, but not veins****
- Some CCBs (DHP) cause reflex tachycardia → $\uparrow$ myocardial O₂ demand

Relative cardiovascular effects of CCBs



Compound	Coronary vasodilation	Suppression of cardiac contractility	Suppression of SA node	Suppression of AV node
Verapamil	++++	++++	+++++	+++++
Diltiazem	+++	++	+++++	++++
Nifedipine	+++++	+	+	0
Nicardipine	+++++	0	+	0

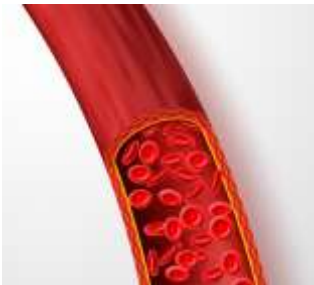
Vascular effects: DHP > Non-DHP, Cardiac effects: Non-DHP >>> DHP

CCBs: Clinical effects

Vascular smooth muscle & Cardiac cells

Effects on vascular smooth muscle

- **Inhibit L- and/or T-type Ca^{2+} channels**
- **“Vascular selectivity”**
 - **Relax arterial smooth muscle -> ↓ afterload (little or no effect on venous)**
 - Specific DHP exhibit greater potencies in some vascular (more selectivity)
 - Nimodipine: Cerebral blood vessels
 - Nicardipine: Coronary vessels



Effects on cardiac cells

- **Negative inotropic effect (myocardial L-type channels)**
 - **Non-DHP:** ↓ inward Ca^{2+} during action potential plateau phase
 - **DHP:** Little effect
- **Negative chronotropic effect (L- and T-type channels)**
 - **Non-DHP:** Depress SA and AV conduction
 - **DHP:** Little effect (but can cause reflex tachycardia)



CCBs: Adverse effects & Cautions

- **Adverse effects:**
 - Major:
 - ↓ contractility
 - AV block, bradycardia, cardiac arrest
 - Minor: hypotension, dizziness, edema, flushing, constipation
- **Drug interactions:**
 - **β-blockers + Non-DHP**:**
Bradycardia, AV block (additive negative inotropic & negative chronotropic effects)
 - Some CCBs (verapamil, diltiazem) cause an increase in plasma digoxin levels -> AV block
- **Contraindication:**
 - **Non-DHPs**
 - SA: digitalis toxicity, β-blocker, SA block
 - AV: digitalis toxicity, β-blocker, AV block
 - **DHPs**
 - Aortic stenosis

CCBs: Clinical uses

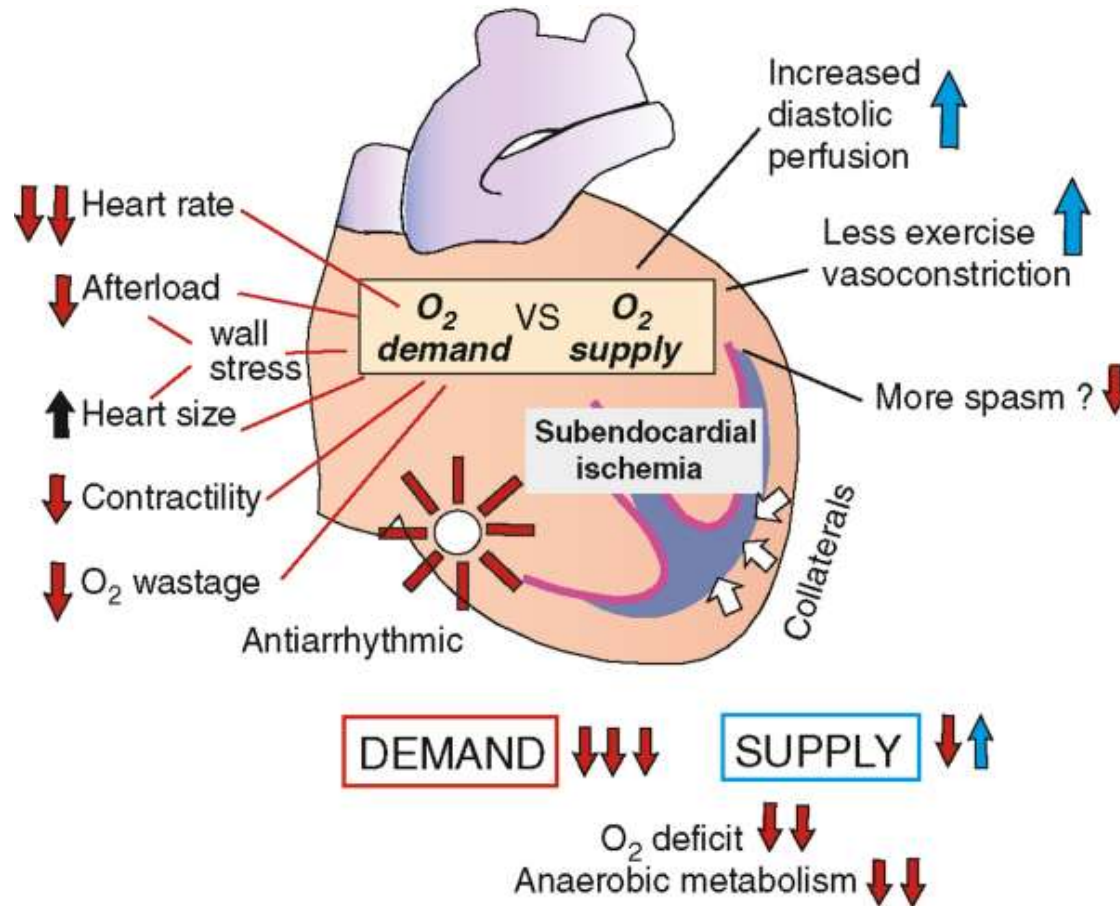
Clinical uses: Stable angina, unstable angina, variant angina

Not a first-line drug, use in patients **unresponsive with β -blockers**

Drug	Oral F (%)	Half-life (hours)	Indication	Dosage
<u>Dihydropyridines</u>				
Amlodipine	65–90	30-50	Angina, hypertension	5–10 mg orally once daily
Felodipine	15-20	11-16	Hypertension, Raynaud's phenomenon	5–10 mg orally once daily
Nicardipine	35	2-4	Angina, hypertension	20–40 mg orally every 8 hours
Nifedipine	45-70	4	Angina, hypertension, Raynaud's phenomenon	3–10 mcg/kg IV; 20–40 mg orally every 8 hours
Nitrendipine	10-30	5-12	Hypertension	20 mg orally once or twice daily
<u>Non-dihydropyridines</u>				
Diltiazem	40-65	3-4	Angina, hypertension, Raynaud's phenomenon	75–150 mcg/kg IV; 30–80 mg orally every 6 hours
Verapamil	20-35	6	Angina, hypertension, arrhythmias, migraine	75–150 mcg/kg IV; 80–160 mg orally every 8 hours

β -blockers

β -blockers: Clinical effects on angina pectoris



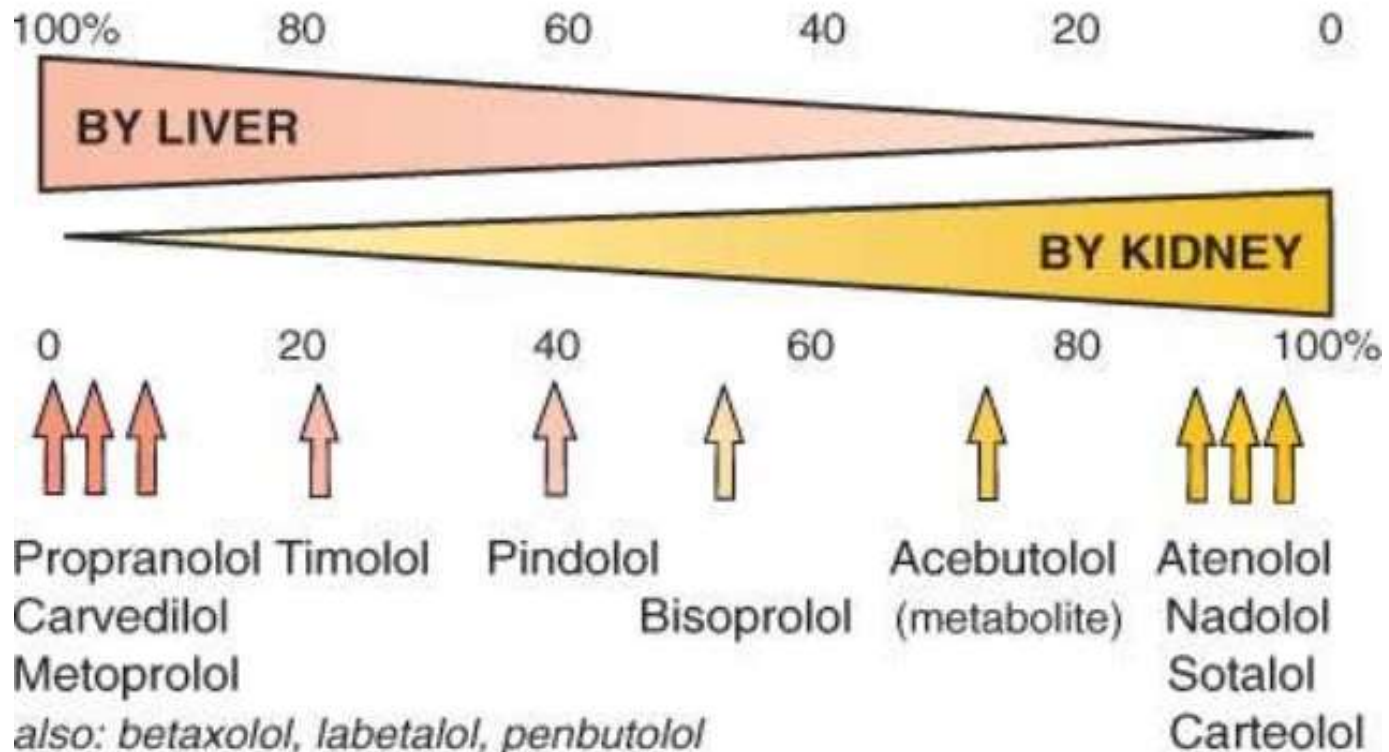
β -Blockers reduces oxygen demand by reducing the double product ($HR \times BP$) and by limiting exercise-induced increases in contractility

β -blockers: Classifications

- **Non-selective β - blockers:**
 - Propranolol, alprenolol, timolol, nadolol
 - Labetalol (ISA, α_1), penbutolol (ISA), pindolol (ISA), carteolol (ISA)
 - carvedilol (α_1 , NO)
- **β_1 - blockers:**
 - Atenolol, esmolol, metoprolol, acebutolol, bisoprolol, acebutolol (ISA), nebivolol (NO)
- **ISA (intrinsic sympathomimetic activity): Relax arterioles**
- **NO: Direct vasodilation**

β -blockers: Pharmacokinetics

Pharmacokinetics: Route of elimination



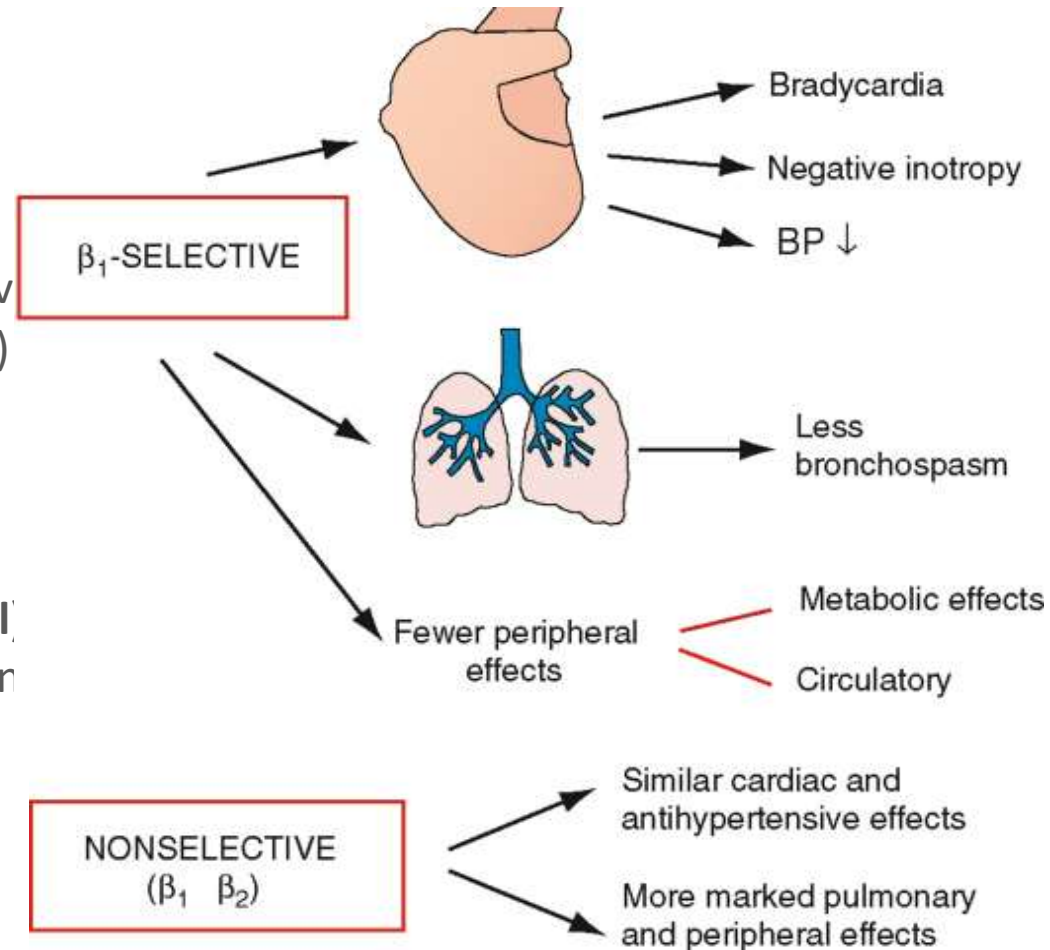
β -blockers: Adverse effects & Cautions

Adverse effects:

- **Bradycardia** (Caution: heart block)
- **Decreased cardiac contractility** (Caution: heart failure)
- **Bronchospasm:** esp. non-selective β -blocker (Caution: asthma, COPD)
- **Impaired glucose tolerance** (Caution: DM)
- **Dyslipidemia** ($\uparrow$ TG, $\downarrow$ HDL)
- **CNS penetration** (esp. propranolol) insomnia, depression, hallucination
- **Masked hypoglycemic symptoms**

Contraindication:

- **Variant angina**
- Sinus bradycardia, heart block
- Asthma, severe COPD

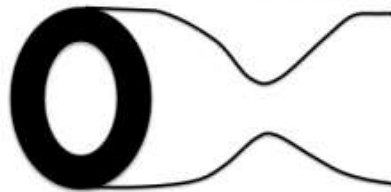


β -blockers: Clinical uses

- Clinical uses:
 - Treatment: Post-MI, stable angina, unstable angina
 - Should **not use in variant angina**: **non-selective β -blockers** (β_2 -AR-blocking effect at coronary vessels $\rightarrow$ induce vasospasm)
 - Should **not use in severe angina**: **β -blockers with ISA activity** (e.g., carteolol, pindolol, acetabutolol) **in severe angina**
 - Partial agonist: $\downarrow$ contractility & HR less than full antagonist

Beta Blockers & Coronary Flow in Variant Angina

β_2 : dilate α : constrict

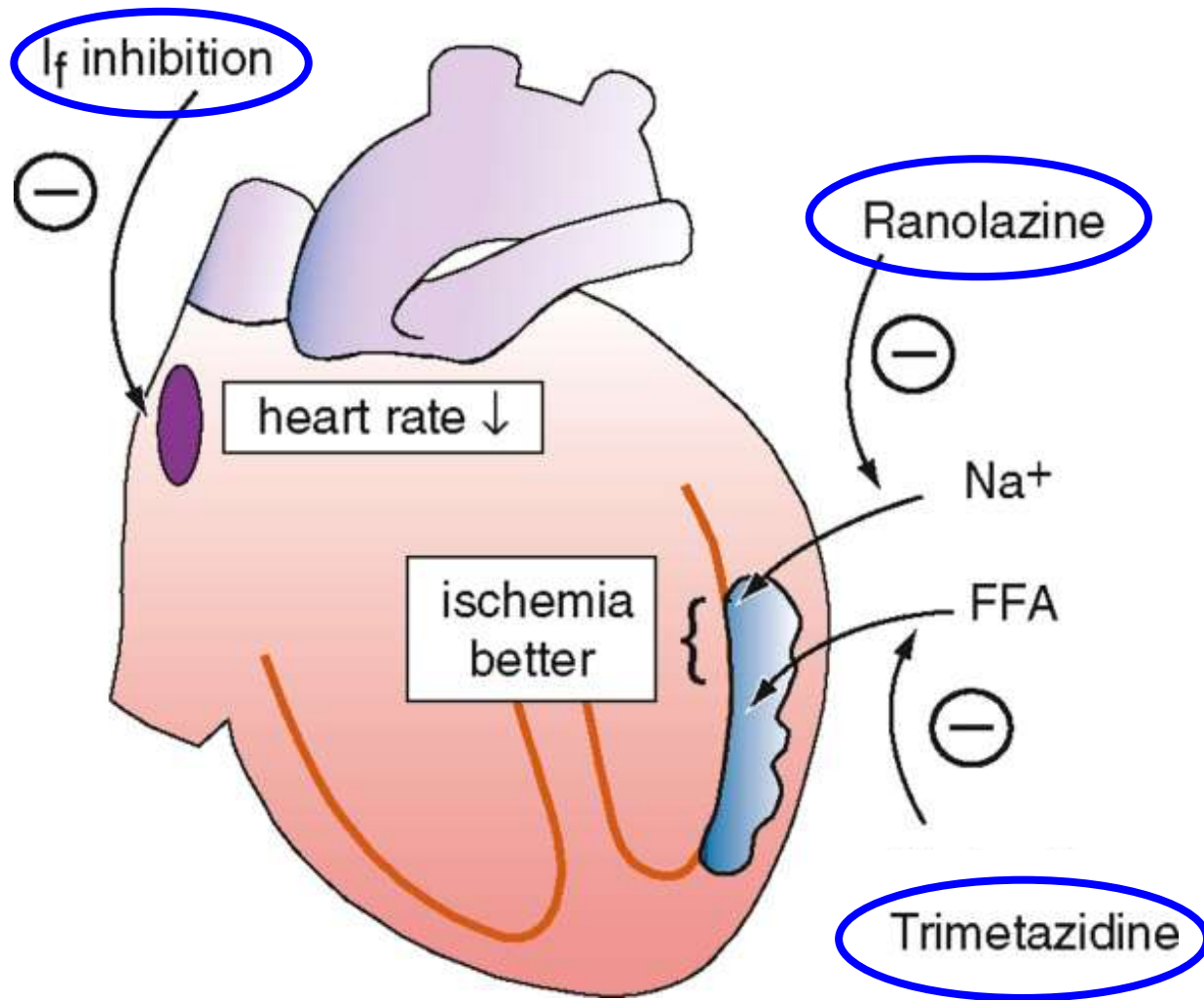


Drugs that block β_2 receptors will leave
vasoconstricting α receptors unopposed
(potentially making angina worse)

Other antianginal drugs



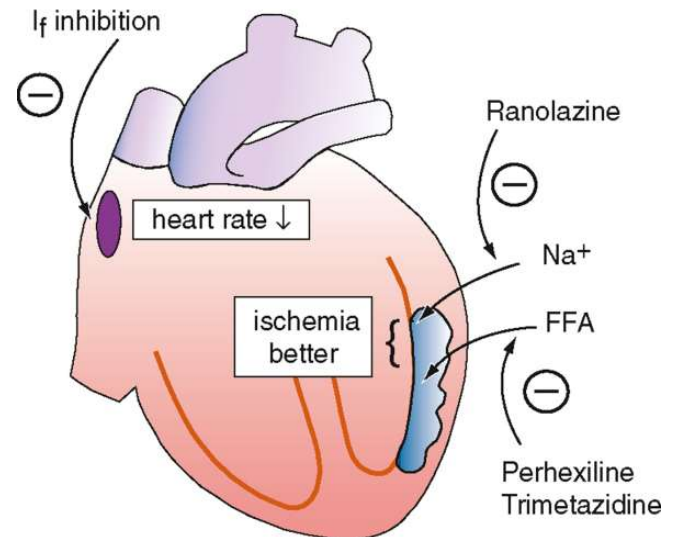
Other antianginal agents work in different ways



Direct bradycardic agents

Ivabradine (Coralan®)

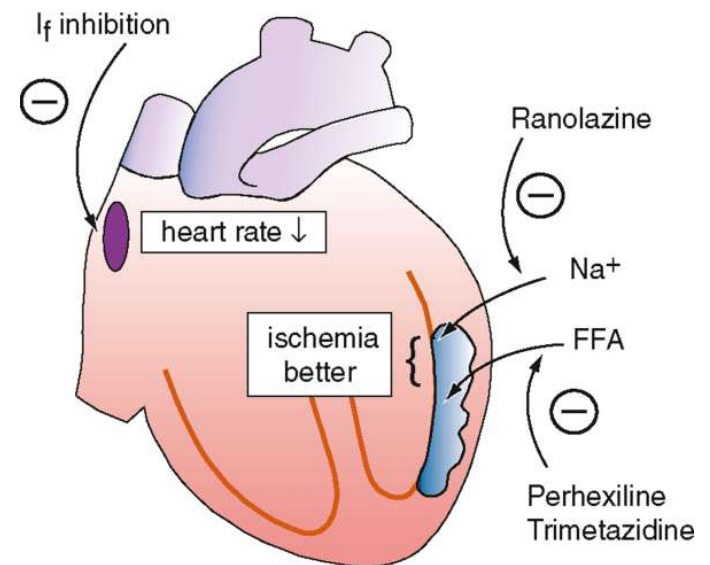
- **Mechanism:**
 - **Selective I_f sodium channel blockers**
 - ↓ Heart rate by inhibiting Na^+ channel in SA node
- **Clinical effects:**
 - ↓ Anginal attack (similar efficacy to CCBs & β blockers)
 - Advantage: Lack of effect on GI and bronchial smooth muscle
- Use in angina when **β -blockers are not tolerated or are contraindicated**



Inhibitors of slowly inactivating sodium current

Ranolazine (Ranexa®)

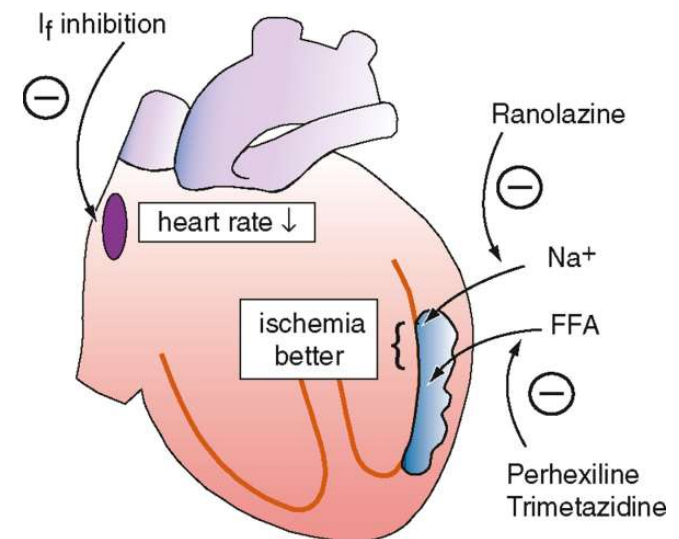
- **Mechanism:**
 - **Inhibition of slow inward Na^+ current (I_{Na})** $\rightarrow$ $\downarrow$ cardiac contractility
- Approved by the U.S. FDA for angina
- Use **only when no response to other antianginals because of the risk of QT prolongation**
- **Caution:**
 - **QT prolongation****
 - CYP3A4 inducer/inhibitor
 - Chronic liver disease



Metabolic modulators

Trimetazidine (Vastarel MR®)

- **Mechanism:**
 - **pFOX inhibitors: Partially inhibit fatty acid oxidation in myocardium**
- **Clinical effects:**
 - In ischemic myocardium, metabolism shifts to oxidation of fatty acids -> ↑ oxygen requirement per unit of ATP produced
 - Partial inhibition of the enzyme required for fatty acid oxidation improve metabolic status of ischemic tissue
- Trimetazidine has demonstrated efficacy in **stable angina**



Combination therapy of angina

Use of more than one class of antianginal agent can **reduce specific undesirable effects of single agent therapy**

Effect	Nitrates	β -blockers or CCB alone	Nitrates <u>plus</u> β -blocker <u>or</u> CCB
Heart Rate	<i>Reflex Increase</i>	Decrease*	Decrease
Afterload	Decrease	Decrease	Decrease
Preload	Decrease	<i>Increase</i>	None or decrease
Contractility	<i>Reflex increase</i>	Decrease*	None
Ejection time	Decrease	<i>Increase</i>	None

Undesirable effects are shown in italics

* DHP may cause the opposite effect due to a reflex increase in sympathetic tone

Combination therapy of angina



Recommended combinations:

- Nitrates + β -blockers
- Nitrates + Non-DHP CCBs
(in patient who cannot use β -blockers)
- Nitrates + β -blockers + CCBs



Non-recommended combinations:

- Nitrates + DHP CCBs (_dipine)
- β -blockers + Non-DHP CCBs

Case study



- A 52-year-old man presents with a history of recent onset chest discomfort when jogging or swimming vigorously. The pain is substernal and radiates to his jaw but disappears after 10-15 minutes of rest.
- He has a history of hyperlipidemia (total cholesterol 245 mg/dL, LDL 160 mg/dL) and admits that he has not been following the recommended diet. His father survived a “heart attack” at age 55 and an uncle died of some cardiac disease at age 60.
- On physical examination, the patient’s blood pressure is 145/95 mm Hg and his heart rate is 80 bpm. There are no other significant physical findings and an electrocardiogram is normal except for slight left ventricular hypertrophy.

Case study

- Assuming diagnosis?
- Risk factor?
- Treatment of acute angina?
- To prevent episodes of angina?
- Long-term prevention?

